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A lightweight CNN-Transformer network for pixel-based crop mapping using time-series Sentinel-2 imagery

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ABSTRACT

Deep learning approaches have provided state-of-the-art performance in crop mapping. Recently, several studies have combined the strengths of two dominant deep learning architectures, Convolutional Neural Networks (CNNs) and Transformers, to classify crops using remote sensing images. Despite their success, many of these models utilize patch-based methods that require extensive data labeling, as each sample contains multiple pixels with corresponding labels. This leads to higher costs in data preparation and processing. Moreover, previous methods rarely considered the impact of missing values caused by clouds and no-observations in remote sensing data. Therefore, this study proposes a lightweight multi-stage CNN-Transformer network (MCTNet) for pixel-based crop mapping using time-series Sentinel-2 imagery. MCTNet consists of several successive modules, each containing a CNN sub-module and a Transformer sub-module to extract important features from the images, respectively. An attention-based learnable positional encoding (ALPE) module is designed in the Transformer sub-module to capture the complex temporal relations in the time-series data with different missing rates. Arkansas and California in the U.S. are selected to evaluate the model. Experimental results show that the MCTNet has a lightweight advantage with the fewest parameters and memory usage while achieving the superior performance compared to eight advanced models. Specifically, MCTNet obtained an overall accuracy (OA) of 0.968, a kappa coefficient (Kappa) of 0.951, and a macro-averaged F1 score (F1) of 0.933 in Arkansas, and an OA of 0.852, a Kappa of 0.806, and an F1 score of 0.829 in California. The results highlight the importance of each component of the model, particularly the ALPE module, which enhanced the Kappa of MCTNet by 4.2% in Arkansas and improved the model's robustness to missing values in remote sensing data. Additionally, visualization results demonstrated that the features extracted from CNN and Transformer sub-modules are complementary, explaining the effectiveness of the MCTNet.

1. Introduction

Reliable crop maps are essential for monitoring agricultural land use and aiding decision-makers in issuing effective policies to guide sustainable agricultural development (Bargiel, 2017). Remote sensing has become the primary means of crop mapping due to its extensive range of observation, regular revisit time, and low cost. Especially, after the successful launch of Sentinel-2 satellites, it is easier to access optical data with fine temporal and spatial resolutions. The time-series optical imagery characterizes the dynamic growth patterns of crops, providing additional valuable information for crop classification (Ashourloo et al., 2019; Yuan et al., 2022).

Traditional crop mapping approaches using time-series remote sensing data can be broadly categorized into three types: threshold-based, statistics-based, and machine learning methods. Threshold-based methods primarily consider the optimal seasonal threshold to separate different crop types based on phenological characteristics and expert experience (Chen et al., 2023; Huang et al., 2022). Statistics-based approaches focus on statistically linking phenology and biophysical changes during crop growth by constructing predefined mathematical functions or models (Achanccaray et al., 2017; Asad and Bais, 2020). However, the aforementioned approaches have limited modeling

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capacity and can be affected by inter-annual and regional discrepancies (Blickensdörfer et al., 2022; Qiu et al., 2018). Machine learning has the advantage of effectively utilizing a wider range of features. Various machine learning models have been successfully applied to crop mapping, such as support vector machine (SVM) and random forest (RF) (Feng et al., 2019; Maponya et al., 2020). However, several drawbacks exist in traditional machine learning models. First, feature engineering is heavily relied upon to select important information and remove redundant information (Xu et al., 2020). Second, time-series spectral features are often treated as a collection of independent features, ignoring the temporal correlations (Inglada et al., 2017; Interdonato et al., 2019).

With the fast development of artificial intelligence, deep learning models have exhibited immense potential in crop mapping (Zhong et al., 2019). Superior to traditional machine learning methods, deep learning can automatically learn discriminating features from complex information, eliminating the effort to select features manually (Schmidhuber, 2015). Among the various deep learning approaches, convolutional neural networks (CNNs), recurrent neural networks (RNNs), and Transformers have been widely used to classify crops (Rußwurm and Körner, 2020). CNNs apply convolution operations to extract features. Since convolution has a fixed kernel size which leads to a local receptive field, CNNs are highly efficient at extracting local information but have insufficient ability to capture global information (Wang et al., 2022a). RNNs are specifically designed for time-series data modeling, which sequentially processes each time step through recurrent cells. In particular, long short-term memory (LSTM) is a widely used RNN method for land cover classifications using time-series images (Sun et al., 2019). Unlike RNNs, Transformers utilize the multi-head self-attention mechanism and positional encoding to process all time steps of sequential data simultaneously (Vaswani et al., 2017). This mechanism is inherently flexible and can understand the relationships between distant parts of the input sequence to obtain both local and global information. Different from the fixed receptive field of CNNs, Transformers can adapt their attention dynamically and capture long-range dependencies in sequential data, which is essential for understanding the overall temporal dynamics and growth patterns of crops across an entire season. For example, Yuan and Lin (2021) proposed a Transformer-based model for crop mapping and achieved better performance than RNNs. Additionally, several studies have proposed more efficient models based on Transformers for mining time-series images, such as temporal attention encoder (TAE) (Garnot et al., 2020) and lightweight temporal attention encoder (LTAE) (Garnot and Landrieu, 2020).

Since CNNs excel at extracting local features while Transformers are adept at capturing long-term dependencies, several recent studies have combined the strengths of both approaches to enhance crop mapping performance (Liu et al., 2022; Niu et al., 2022; Tang et al., 2024; Wang et al., 2022a; Xiang et al., 2023). However, these hybrid models have primarily focused on patch-based crop mapping in small regions. Although patch-based models can utilize additional spatial information compared to pixel-based models, the cost of patch-based sampling is more expensive (Feng et al., 2023), limiting their application in large areas. Moreover, most of these models used an encoder-decoder structure commonly applied in language translation and segmentation tasks. The encoder-decoder structures can increase the volume of the model, making it unsuitable for lightweight applications. In contrast, pixel-based crop mapping is a typical classification problem that only requires an encoder to extract features for classification (Li et al., 2020). Zhang et al. (2023) integrated a CNN and a modified Transformer to design a global–local temporal attention encoder (GL-TAE) for pixel-based crop mapping. In their approach, time-series multi-spectral data is transformed into a 2D matrix, enabling the CNN to extract local temporal patterns. These local features are then merged with features extracted by the Transformer to enhance crop mapping. The results demonstrate that this hybrid model significantly outperforms both the CNN and the Transformer models.

However, previous CNN-Transformer models simply integrated features extracted from CNN and Transformer while giving limited consideration to missing data. Missing values in time-series images are often inevitable due to factors such as cloud cover and non-observation. These missing values can disrupt the inherent temporal patterns of crops, potentially affecting the performance of models, especially those that consider sequential information, such as Transformers. Although Rußwurm and Körner (2020) found that the self-attention mechanism of Transformers reduces the impact of missing values in time-series remote sensing data for crop mapping to a certain extent, it is not specifically designed to address missing values, and thus performance improvements are necessary. Several studies have applied interpolation techniques to eliminate missing values (Wang et al., 2022b; You et al., 2021). However, performing interpolation demands extensive computational resources for multi-bands in large temporal-spatial scales. Moreover, there remains an inherent uncertainty regarding the fidelity of the reconstructed data to the actual values, which raises concerns about the reliability of interpolation in applications where the accuracy of temporal information is paramount.

Therefore, we propose a lightweight multi-stage CNN-Transformer network (MCTNet) for pixel-based crop mapping using time-series sentinel-2 imagery. MCTNet consists of several successive CNN-Transformer fusion (CTFusion) modules with pooling operations and a multilayer perceptron (MLP) classifier. Each CTFusion has a CNN sub-module and a Transformer sub-module. The CNN sub-module is designed with a 1D convolutional network to extract local spectral information along the time dimension. The Transformer sub-module is designed based on the encoder of the original Transformer to capture the long-range dependencies of the time series data and extract the features from a global perspective. Besides, we designed an attention-based learnable positional encoding module (ALPE) in the first Transformer sub-module to capture the temporal correlation in the time series data under different missing rates. We selected California and Arkansas in the U.S. as study areas to verify the effectiveness of the model with distinct planting structures. At the same time, eight advanced models were selected for comparative verification. The main contributions of our research are:

(1) We propose the MCTNet that integrates the advantages of the CNN and Transformer with a lightweight architecture for pixel-based crop mapping.

(2) ALPE is designed to reduce the impact of missing values and provide reasonable positional encoding, which can improve the model's ability to handle time-series data under complex missing conditions.

(3) Compared with various state-of-the-art models, the proposed model exhibits superior performance in key areas, including parameter size, classification accuracy, and the quality of mapping results.

2. Materials and methods

2.1. Study area

Two states in the U.S. were chosen as the study areas, including California and Arkansas (Fig. 1 (a)). California has an extremely diverse natural landscape, including grasslands, deserts, rainforests, mountains, and oceans. The agricultural system in this state is quite complex, with small plots and a variety of crop types (Zhong et al., 2011). Arkansas is an important agricultural state in the U.S. with over 40 % of the land dedicated to agricultural production (Martin et al., 2021). It is a major producer of crops such as rice, soybeans, corn, cotton, and wheat, which are extensively cultivated on large farms. Due to the differences in major crops and cultivation methods between California and Arkansas, the crop mapping models can be thoroughly validated. Besides, two distinct mapping areas were selected in each state to assess the practical application capability of the models (Fig. 1 (b) and Fig. 1 (c)). These areas were selected for their considerable geographical separation and

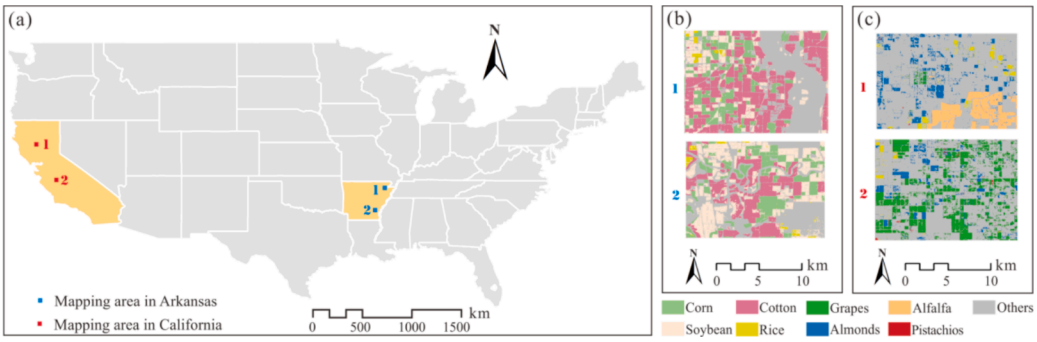

Fig. 1. Study area. (a) Arkansas and California. (b) Mapping areas in Arkansas. (c) Mapping areas in California (The information on crop types is from CDL).

comprehensive coverage of various crop types, which allows for better validation of the model's validity.

2.2. Datasets

Three types of datasets were collected in this study, including ground truth of crop type, land cover data, and remote sensing images. Table 1 provides a summary for each data source and the detailed descriptions are in the following sections.

2.2.1. Ground truth

Cropland Data Layer Data (CDL) is one of the most widely used crop maps published by the U.S. Department of Agriculture (USDA). CDL describes the distribution of more than 100 crops in the U.S., completely covering Arkansas and California. It has an annual update frequency, high spatial resolution (30 m), and high classification accuracy (Boryan et al., 2011). Moreover, it offers a confidence layer that indicates the predicted confidence level for each pixel. Therefore, this study used the CDL data to generate labeled samples for model training and testing.

Table 1
A summary of the data used in this study.

Category	Data Name	Variable Name	Spatial Resolution	Temporal Resolution	Source
Ground Truth of Crop Type	Crop Data Layer	Crop type	30 m	Yearly	USDA
Land Cover Map	ESA WorldCover 2021	Land cover	10 m	–	ESA
Remote sensing images	Sentinel-2 Level-2A	Band 2 Blue	10 m	5-day	ESA
		Band 3 Green	10 m		
		Band 4 Red	10 m		
		Band 5Red Edge-1	20 m		
		Band 6 Red Edge-2	20 m		
		Band 7 Red Edge-3	20 m		
		Band 8 NIR	10 m		
		Band 8A Red Edge-4	20 m		
		Band 11 SWIR 1	20 m		
		Band 12 SWIR 2	20 m		

2.2.2. ESA WorldCover 2021

The ESA WorldCover 2021 is a comprehensive global land cover map for 2021, offering a fine spatial resolution of 10 m. This pioneering product, derived from Sentinel-1 and Sentinel-2 data, provides a detailed classification of the earth's surface into 11 land cover classes. Since this study focuses on crop mapping, ESA WorldCover 2021 was used to mask non-cropland areas, thereby streamlining sample collection and practical mapping applications.

2.2.3. Remote sensing images

Sentinel-2 is a multi-spectral imaging system provided by the European Space Agency (ESA) and designed for high-resolution imaging of the Earth's surface. It comprises two satellites that can be combined to provide full coverage of the Earth's surface every five days. Sentinel-2 has 13 bands ranging from the visible to the shortwave infrared, with spatial resolutions varying up to a maximum of 10 m (Table 1). Among these bands, Band 1, Band 9, and Band 10 have an inadequate resolution (60 m) and contribute less to the fine classification of crops; thus, we selected the remaining ten bands as spectral features, including three visible bands, one near-infrared (NIR) band, four red-edge bands, and two shortwave infrared (SWIR) bands. Besides, Sentinel-2 Level-2A surface reflectance images (bottom-of-atmosphere) were applied in this study because these images are atmospherically corrected and orthorectified, contributing to more accurate and reliable monitoring of agricultural conditions.

2.2.4. Data preprocessing

For remote sensing observations, we collected all Sentinel-2 images over the study areas from the year 2021. Subsequently, we eliminated cloud-affected pixels and computed the median value of the remaining observations at ten-day intervals, resulting in a total of 36 temporal sequences. It is noted that there are still missing data in the sequences, and we used 0 to mark the missing data. We used the CDL map from 2021 to collect labeled crop samples. In particular, we set a 95 % confidence to filter the CDL map to improve the quality of sampling and used the ESA WorldCover 2021 to mask non-cropland areas. Then, we randomly sampled 10,000 points in each study area and extracted the corresponding crop type from the CDL map, as well as the associated spectral features from the time-series images. As a result, each labeled sample has 360 spectral features, consisting of 10 spectral bands with 36 temporal observations. The distribution of the crop samples is shown in Table 2. Specifically, crop types that constitute less than 5 % of the total number of samples were merged into a category labeled as "others". The NDVI time-series curve of each crop is generated by averaging the NDVI values of all the corresponding crop samples (Fig. 2). All the necessary steps for data collection and preprocessing were executed using the Google Earth Engine.

2.3. Multi-stage CNN-Transformer network (MCTNet)

The primary objective of the proposed MCTNet is to integrate the

Table 2
Numbers of the samples in the study areas.

Study area	Class Name	Number of Samples	Training	Validation	Testing
Arkansas	Soybeans	4677	240	60	4377
	Rice	2423	240	60	2123
	Corn	1522	240	60	1222
	Cotton	762	240	60	462
	Others	616	240	60	316
	All	10,000	1200	300	8500
California	Grapes	2054	240	60	1754
	Rice	2037	240	60	1737
	Alfalfa	974	240	60	674
	Almonds	783	240	60	483
	Pistachios	640	240	60	340
	Others	3512	240	60	3212
	All	10,000	1440	360	8200

strengths of CNNs and Transformers for pixel-based crop mapping. CNNs are particularly effective at extracting local features while Transformers are proficient at capturing long-term dependencies. By combining these capabilities, MCTNet is expected to construct a more comprehensive representation of crop growth patterns, thereby enhancing the accuracy of crop mapping. Additionally, the ALPE module is specifically designed to handle missing data effectively, ensuring both robustness and accuracy of the model even in the presence of incomplete data. The MCTNet uses several successive CTFusion modules with pooling operations to extract the multi-level local and global information of the time-series data. The structure of the proposed MCTNet with three stages is shown in Fig. 3. Given time-series images, a pixel sample can obtain time-series multi-spectral features (Input 1) and a set of time-series data indicating whether the current temporal data is missing (Input 2). Input 1 is converted to a 2D matrix for model processing. Specifically, the time-series features are extracted by the CTFusion modules sequentially and then are fed to an MLP classifier to predict the crop type of the sample. A CTFusion is composed of two parallel components, a Transformer sub-module, and a CNN sub-module. In particular, the traditional positional encoding was replaced by an ALPE module in the first Transformer sub-module. The ALPE module utilizes Input 2 to generate an enhanced positional representation. A detailed description of the modules in the proposed model is provided in the following.

2.3.1. *Transformer sub-module*

The Transformer sub-module is designed based on the encoder of the original Transformer to capture the contextual information of the time-series data. The architecture of the sub-module is shown in Fig. 4. It removes the input embedding of the original Transformer and mainly consists of two key components: multi-head self-attention and positional encoding.

The self-attention component computes attention scores for each temporal feature, enabling the model to focus on relevant context while suppressing noise. Multi-head self-attention utilizes H sets of learned attention weights, permitting the model to capture diverse relationships and dependencies within the data. The input of the self-attention component consists of query (Q), key (K), and value (V) which are calculated by the module input and three different transformation matrices. Q and K are used to calculate an attention weight matrix $W \in [0,1]^{T\times T}$ (where T is the number of time steps), and the output of self-attention is produced by W and V:

$$Attention(Q,K,V) = WV = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where d_k is the column number of the Q.

Positional information is crucial in sequential data. In the original Transformer, absolute positional encoding using a fixed sinusoidal encoding mechanism with a predefined wavelength is used to represent the order of the temporal input (Vaswani et al., 2017). The encoded position can be calculated as:

$$PE(t_i)_p = \begin{cases} sin(t_i/10000^{2k/d_m}), ifp = 2k \\ \cos(t_i/10000^{2k/d_m}), ifp = 2k+1 \end{cases} \quad (2)$$

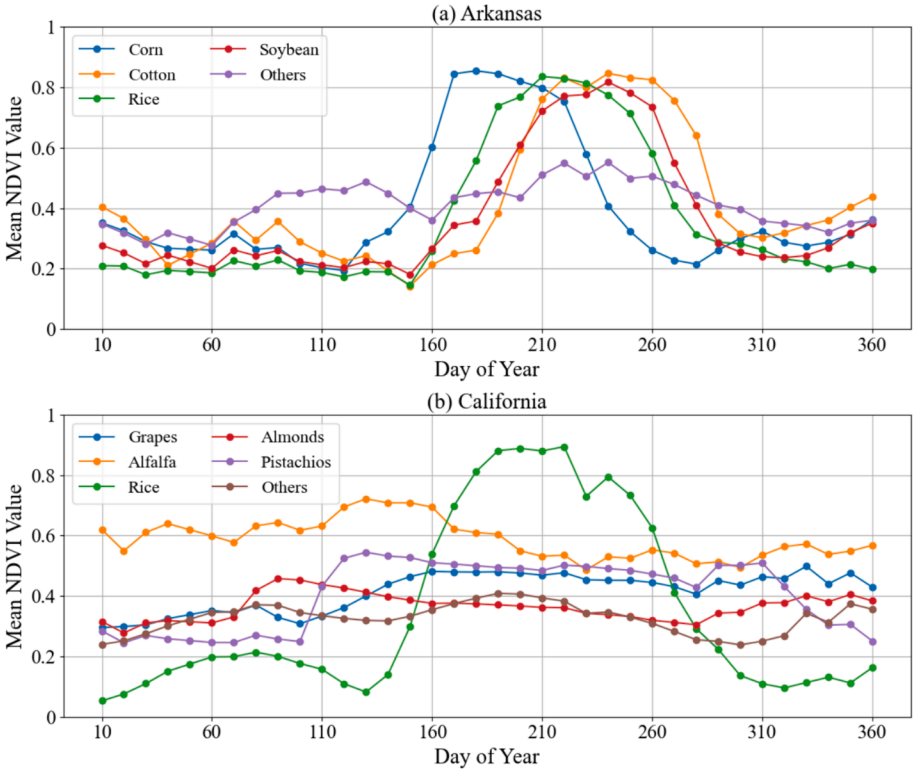

Fig. 2. NDVI time-series profiles of the crops in (a) Arkansas and (b) California.

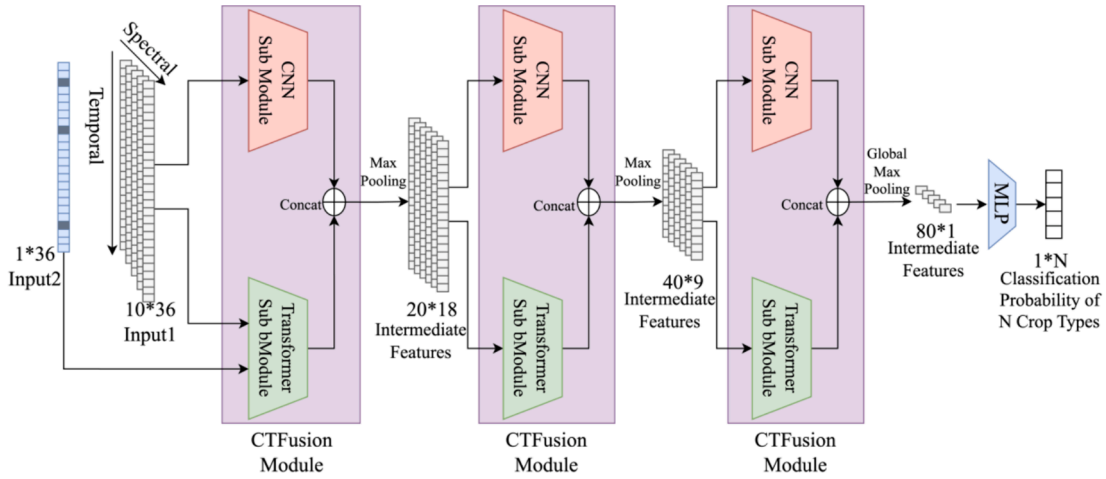

Fig. 3. Architecture of the MCTNet with three stages.

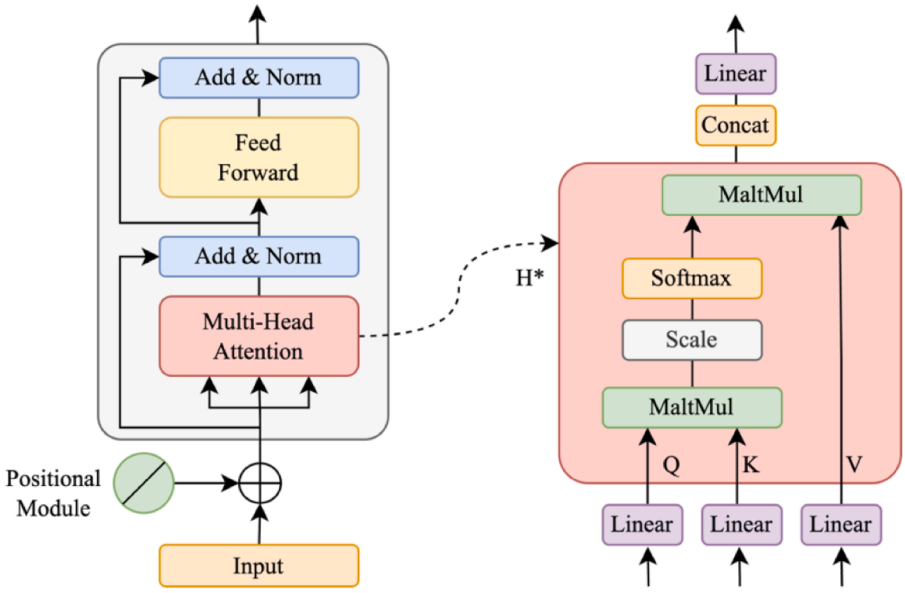

Fig. 4. Architecture of the Transformer sub-module.

where t_i is the temporal order, and d_m is the encoding size.

The absolute positional encoding is added to the input to enable the model to attend to both content and position. However, the absolute positions lack flexibility and treat all positions equally regardless of the contextual information. Besides, gaps often occur in the time-series data due to clouds, and the absolute positional encoding cannot help the model understand which temporal data is unreliable. Therefore, we designed an attention-based learnable positional encoding (ALPE) module (Fig. 5) to solve the above issues. This module first uses absolute positional encoding as the initial positional vector to transform the time feature from one dimension to two dimensions to meet the input format. The two-dimensional positional vector is masked by Input 2 (mentioned in Section 2.3) and then processed by a 1D convolution layer. It is worth noting that the mask is only used in the first stage of the model. After that, the Efficient Channel Attention (ECA) module is applied to the

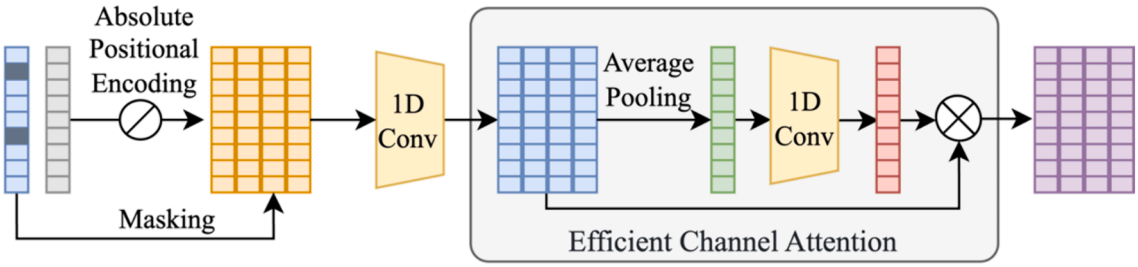

Fig. 5. Attention-based learnable positional encoding module.

processed positional vector. ECA is an efficient attention module that has few parameters and can bring clear performance gain in 1D CNNs (Wang et al., 2020). The final position matrix is expected to learn the complex relationships from the time-series features and the importance of different temporal positions under different missing rates.

$$ALPE(t) = ECA(Conv1D(PE(t) \times mask)) \quad (3)$$

where t represents the time series. For example, given a sequential input with 36 time steps and 10 features. The absolute positional encoding of its time series t is PE(t) $\in \mathbb{R}^{36\times10}$ which is a two-dimensional vector. The positions of the missing values in PE(t) are masked by zero to reduce the impact of missing values on the subsequent position encoding learning. Then, the positional vector is processed by a convolution layer and ECA module to form the final positional vector $ALPE(t) \in \mathbb{R}^{36\times10}$.

ALPE incorporates learnable parameters into the standard positional encoding, allowing the encoding to adapt based on the input data. These parameters are trained to minimize the negative impact of missing values, effectively modifying the positional information to better represent the temporal structure of the data. By refining the positional encoding, the ALPE module improves the model's ability to capture the relative relationships between time steps, even in the presence of missing data. This enhanced representation allows the attention mechanism to distribute attention weights more accurately, leading to better information propagation across time steps and increased robustness to missing values.

2.3.2. CNN sub-module and MLP classifier

The CNN sub-module is designed with a very simple structure (Fig. 6). It consists of two consecutive 1D convolution layers that operate along the time dimension of the input. Each convolution layer is followed by a batch normalization layer. The output of the two layers is processed by a ReLU activation function. Besides, a connection directly skips over the two convolutional layers and adds the input of the sub-module to the output of the two layers. This shortcut connection can enable the gradient to flow more easily during training and helps to avoid the vanishing gradient problem. The MLP classifier is made up of a linear layer with a Softmax activation. The input of the MLP is the extracted features from the last CTFusion module. Specifically, the output of the last CTFusion module is processed by a global max pooling operation along the channel dimension to transform its form into a one-dimensional vector. Then the vector is fed to the MLP classifier to predict the crop type.

2.4. Experimental setting

There are 10,000 samples in Arkansas and California, respectively. For each state, the samples were split into training, validation, and testing sets. Specifically, 300 samples per crop type were randomly selected to compose the training and validation datasets (with a rate of 8:2), and the rest samples were used for testing. The details of the sample partition are shown in Table 2. The training samples were used to train the models, and the validation samples were used for selecting hyper-parameters, including the number of stages (n_stage), the number of heads (n_head) in the Transformer sub-module, the kernel size of the CNN sub-module, the learning rate (lr), and the optimizer. To fully assess the effectiveness of the MCTNet, we compared it with eight advanced models, including SVM, RF, LSTM, Transformer, Residual Neural Network (ResNet), TAE, LTAE, and GL-TAE. Particularly, ResNet is a deep CNN with many variants, and this study adopted the ResNet-18, which has 18 convolutional layers. Since the original Transformer is not suitable for classification tasks, this study adopted the modified Transformer model of Rußwurm and Körner (2020), which only contains the encoder part of the original Transformer.

All models were implemented in a Python environment, in which SVM and RF were constructed with the Scikit-Learn library, and the deep learning models were constructed with the PyTorch library. The models all ran on a workstation with an Intel I7-12900 CPU, 64 GB RAM, and an NVIDIA RTX3090 24G GPU. In the training process, the batch size and number of epochs of the deep learning models were set as 32 and 200, respectively. The hyperparameters of the models for tuning, their candidates, and their final value in each study area are listed in Table 3. The other hyperparameters of the models were selected with default values.

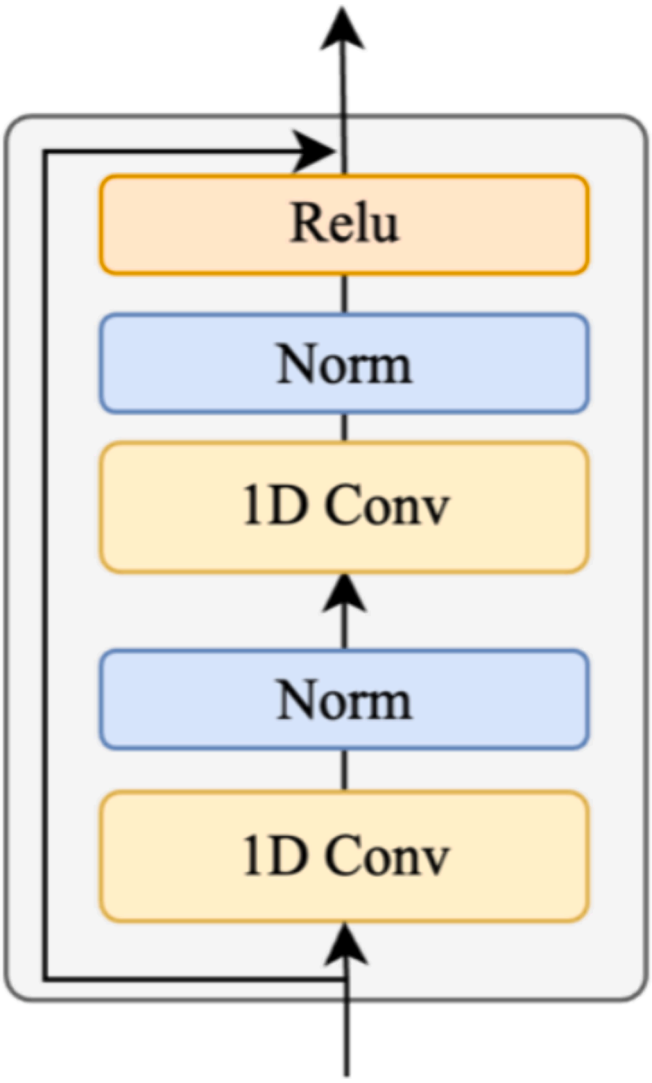

Fig. 6. Architecture of the CNN sub-module.

To quantitatively evaluate the models, the confusion matrix was first computed to show the distribution of all the predicted responses. Then, overall accuracy (OA), kappa coefficient (Kappa), and macro-averaged F1 score were calculated based on the matrix to comprehensively measure the overall performance of the models. In particular, the macro-averaged F1 score represents the average F1 score of all classes, with each F1 score being the harmonic mean of precision and recall for its respective class. The macro-averaged F1 usually serves as a crucial metric for evaluating a model's effectiveness in imbalanced datasets. For the sake of brevity, this metric will be referred to as "F1″ in subsequent discussions. The formulas of the three metrics are shown in Eq. (4)-(6).

$$OA = \frac{1}{N}\sum_{i=1}^{n} x_{ii} \quad (4)$$

$$F1 = \frac{2}{n}\sum_{i=1}^{n}\frac{P_i R_i}{P_i + R_i} \quad (5)$$

Table 3
The hyperparameters of different models in Arkansas and California.

Model	Hyperparameters candidates	Arkansas	California
MCTNet	n_stage = [1, 2, 3,4,5] n_head = [1,2,5,10] kernel size = [1,3,5] lr = [0.0001,0.001,0.01,0.1] optimizer = [SGD, Adam]	n_stage = 3 n_head = 5 kernel size = 3 lr = 0.001 optimizer = Adam	n_stage = 3 n_head = 5 kernel size = 3 lr = 0.001 optimizer = Adam
GL-TAE	d_m = [64,128, 256, 512] n_head = [2,4,8,16,32] lr = [0.0001,0.001,0.01,0.1] optimizer = [SGD, Adam]	d_m = 256; n_head = 8 lr = 0.001 optimizer = Adam	d_m = 256; n_head = 16 lr = 0.001 optimizer = Adam
LTAE	d_m = [64,128, 256, 512] n_head = [2,4,8,16,32] lr = [0.0001,0.001,0.01,0.1] optimizer = [SGD, Adam]	d_m = 256 n_head = 8 lr = 0.001 optimizer = Adam	d_m = 256 n_head = 8 lr = 0.001 optimizer = Adam
TAE	d_m = [64,128, 256, 512] n_head = [2,4,8,16,32] lr = [0.0001,0.001,0.01,0.1] optimizer = [SGD, Adam]	d_m = 128 n_head = 4 lr = 0.001 optimizer = Adam	d_m = 128 n_head = 4 lr = 0.001 optimizer = Adam
Transformer	d_m = [64,128, 256, 512] n_head = [2,4,8,16,32] n_layers = [1,2,4,6,8] lr = [0.0001,0.001,0.01,0.1] optimizer = [SGD, Adam]	d_m = 128 n_head = 4 n_layers = 2 lr = 0.001 optimizer = Adam	d_m = 128 n_head = 4 n_layers = 2 lr = 0.001 optimizer = Adam
Resnet	lr = [0.0001,0.001,0.01,0.1] optimizer = [SGD, Adam]	lr = 0.001 optimizer = Adam	lr = 0.001 optimizer = Adam
LSTM	hidden units = [64,128,256,512] lr = [0.0001,0.001,0.01,0.1] optimizer = [SGD, Adam]	hidden units = 256 lr = 0.001 optimizer = Adam	hidden units = 256 lr = 0.001 optimizer = Adam
RF	n_estimators = [100,200,400] max_depth = [5,10,15,20]	n_estimators = 200 max_depth = 20	n_estimators = 200 max_depth = 20
SVM	C=[0.5, 1, 1.5, 2, 2.5, 3] kernel = [Linear, RBF]	C=1 kernel = RBF	C=1 kernel = RBF

$$\text{Kappa} = \frac{N\sum_{i=1}^{n} x_{ii} - \sum_{i=1}^{n} x_{+i}x_{i+}}{N^2 - \sum_{i=1}^{n} x_{+i}x_{i+}} \quad (6)$$

where N and n represent the number of samples and the number of crop types, respectively. x_{ii} is the ith diagonal value of the confusion matrix. x_{+i} and x_{i+} denote the sum of the predicted samples for class i and the sum of the true samples for class i, respectively. P_i is the precision of class i and R_i represents the recall of class i.

3. Results

3.1. Ablation experiment

To validate the effectiveness of the components, we assessed the impact of the ALPE module, CNN sub-module, and Transformer sub-module on the performance of the MCTNet. Specifically, we removed the three modules separately to build three new comparison models, namely MCTNet_noALPE, MCTNet_noCnn, and MCTNet_noTrans. It is worth noting that the MCTNet_noALPE model used the absolute positional encoding of the original Transformer. Besides, we also removed the data missing mask of the ALPE to construct MCTNet_noMask to assess its effectiveness.

Table 4 shows the overall performance of the models in the two study areas. It is clear that the MCTNet outperformed the other models significantly in both regions, illustrating that each module contributes to improving classification performance. In Arkansas, the MCTNet obtained the best performance with 0.968 OA, 0.951 Kappa, and 0.933 F1, followed by MCTNet_noMask. The MCTNet_noALPE and MCTNet_noCnn performed similarly, with an F1 that is approximately 4 % less than that of MCTNet. The MCTNet_noTrans produced the lowest accuracy, indicating that the Transformer sub-module had the greatest impact on the accuracy of the model. In California, the accuracy of the models decreased significantly compared to the results in Arkansas. This may be attributed to the small crop plots and numerous types of crops in California, which increase the difficulty of classification. The MCTNet still exhibited the best performance, followed by the MCTNet_noMask, MCTNet_noALPE, MCTNet_noCnn, and MCTNet_noTrans.

Table 4
Ablation study of the MCTNet with three evaluation metrics.

Study area	Model	OA	Kappa	F1
Arkansas	MCTNet	**0.968**	**0.951**	**0.933**
	MCTNet_noMask	0.943	0.914	0.895
	MCTNet_noALPE	0.938	0.909	0.888
	MCTNet_noCnn	0.938	0.907	0.876
	MCTNet_noTrans	0.925	0.888	0.860
California	MCTNet	**0.852**	**0.806**	**0.829**
	MCTNet_noMask	0.840	0.792	0.822
	MCTNet_noALPE	0.841	0.792	0.810
	MCTNet_noCnn	0.825	0.773	0.792
	MCTNet_noTrans	0.803	0.744	0.784

3.2. Model performance comparison

The results of the classification accuracy of the nine models in the two regions are displayed in Table 5. A major finding is that the MCTNet achieved the best performance while the LSTM, SVM, and TAE yielded the worst results in both regions. LSTM relies on sequential data to learn temporal dynamics, but missing data can impact its ability to capture temporal dependencies. SVM is also sensitive to noise and outliers in the dataset (Sabzekar and Hasheminejad, 2021). RF yielded significantly superior accuracy compared to LSTM, SVM, and TAE. This is because RF can exhibit robustness through majority voting with its numerous trees. Apart from the proposed MCTNet, the GL-TAE also performed well in both study areas. This performance can be attributed to the fact that both MCTNet and GL-TAE integrate CNN and self-attention architectures. Despite their differing self-attention mechanisms, this integration facilitates the advantageous combination of global and local attention, which is critical for effective crop mapping. Generally, the MCTNet outperformed other comparison models remarkably in all metrics.

To better analyze the classification results of different models, the

Table 5
Classification accuracy of different models.

Study area	Model	OA	Kappa	F1
Arkansas	MCTNet	**0.968**	**0.951**	**0.933**
	GL-LTAE	0.952	0.928	0.905
	LTAE	0.941	0.912	0.884
	TAE	0.911	0.869	0.854
	ResNet	0.945	0.917	0.894
	Transformer	0.956	0.934	0.909
	LSTM	0.907	0.862	0.843
	RF	0.936	0.904	0.887
	SVM	0.910	0.867	0.846
California	MCTNet	**0.852**	**0.806**	**0.829**
	GL-LTAE	0.843	0.796	0.820
	LTAE	0.838	0.788	0.805
	TAE	0.787	0.728	0.767
	ResNet	0.837	0.787	0.821
	Transformer	0.813	0.759	0.789
	LSTM	0.799	0.739	0.766
	RF	0.823	0.770	0.793
	SVM	0.786	0.724	0.762

normalized confusion matrices are shown in Figs. 7 and 8. Obviously, the MCTNet exhibited a notably lower rate of misclassifications compared to other methods. In Arkansas, the MCTNet demonstrated superior performance in accurately classifying corn, soybeans, and rice. It is worth noting that the MCTNet generated very high accuracy on corn (0.991 OA) and soybeans (0.990 OA), while the other models tended to misclassify the two crops to different degrees. According to Fig. 2, the maturity stage of corn is significantly earlier than that of soybeans in Arkansas. The MCTNet may be able to capture this difference, yielding very few misclassified samples for corn and soybeans. Moreover, the other models erroneously classified a large number of rice samples as cotton or others, while the MCTNet maintained a high classification performance. In California, due to the complicated environment and heterogeneous crop categories, the classification accuracy of all models experienced a remarkable decline. Nevertheless, the MCTNet still achieved better classification accuracy in almonds and others compared to other models. The broad range of crop types included in others greatly increases the difficulty of classification, while the proposed model still achieved an increase in OA of 5.4 % compared to the GL-TAE.

The parameter sizes, processing time, and memory of the comparison models are shown in Table 6. Specifically, we conducted inference and training time evaluations 10 times to ensure a robust comparison, adopting the median values as the final results. The training time covers the whole process of model training, and the inference time is the total time to calculate the results of the testing data with the trained model. The MCTNet has the fewest number of parameters and the lowest memory usage. Compared with the GL-TAE, the MCTNet has nearly half the number of parameters. However, the training and inference time of the MCTNet are not superior to those of other models. This may be due to the sequential stages in MCTNet, which affect the efficiency of parallel computing.

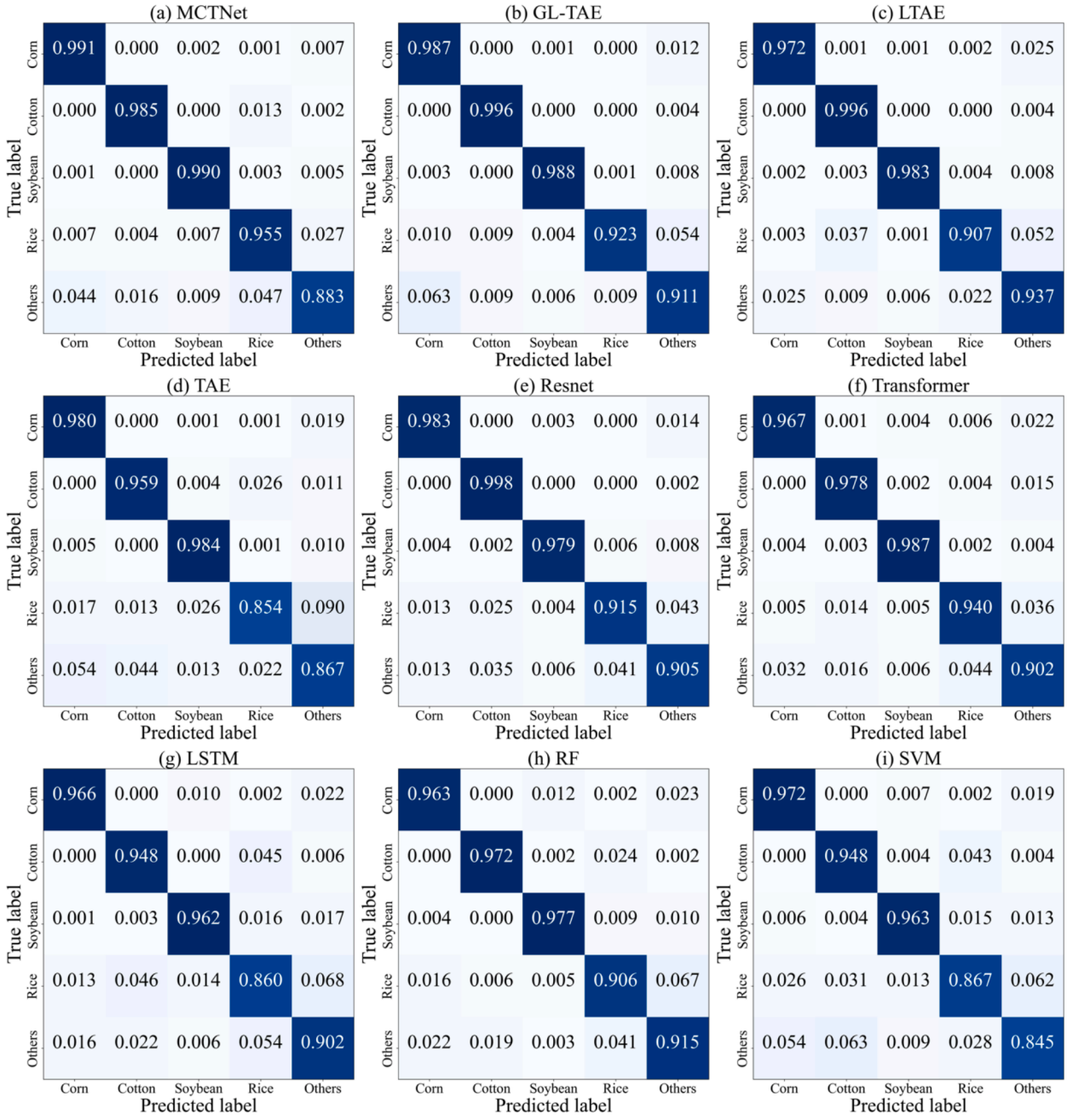

Fig. 7. Confusion matrices of different models in Arkansas.

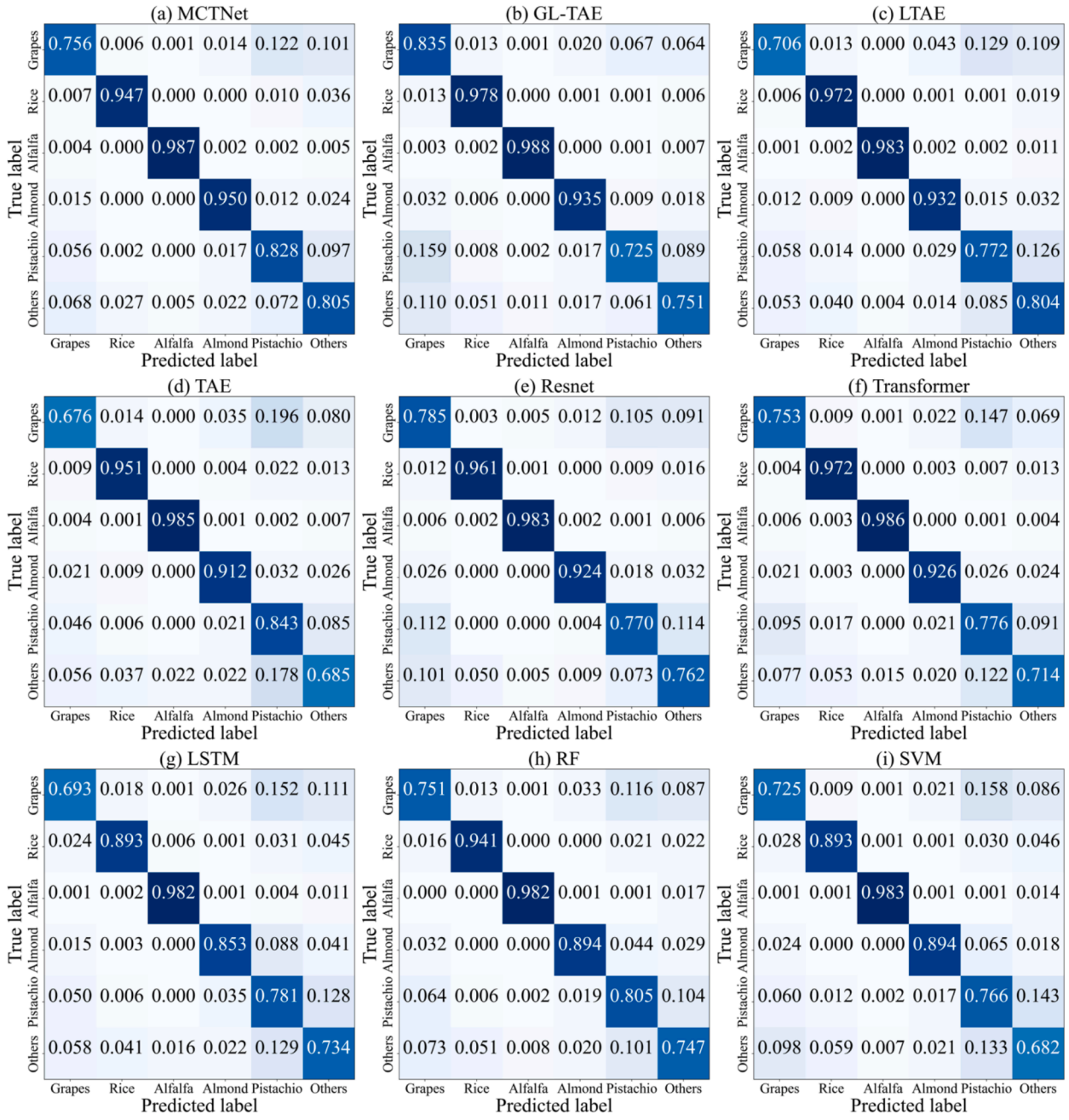

Fig. 8. Confusion matrices of different models in California.

Table 6
The parameter size, processing time, and memory of the crop mapping models.

Model	Arkansas				California			
	Parameter Size	Training (s)	Inference (s)	Memory (MB)	Parameter Size	Training (s)	Inference (s)	Memory (MB)
MCTNet	**55,059**	75.261	1.093	**0.740**	**55,140**	89.191	1.107	**0.740**
GL-TAE	105,289	54.258	0.924	1.306	121,834	70.336	0.967	1.500
LTAE	95,301	50.799	0.902	1.190	95,334	64.366	0.843	1.191
TAE	161,605	53.844	0.918	1.960	161,638	67.053	0.957	1.960
Transformer	1,325,204	88.771	1.106	15.545	1,325,332	103.876	1.198	15.547
Resnet	2,775,237	73.381	0.953	31.800	2,775,494	88.586	0.974	31.800
LSTM	276,741	48.514	**0.716**	3.250	276,998	57.486	**0.736**	3.250
RF	–	**1.530**	1.239	4.969	–	**2.164**	1.267	8.686
SVM	–	2.220	1.339	1.917	–	2.730	1.510	3.026

3.3. Crop mapping performance

The crop mapping results of the models are shown in Fig. 9 and Fig. 10. It should be noted that TAE, LSTM, and SVM were excluded from mapping due to their inferior accuracy. Residual maps were created to illustrate the distribution of errors and correct classifications, with white for correct classifications and black for errors. For quantitative analysis, the mapping accuracy of each residual map was determined by calculating the ratio of correctly classified pixels to the total pixel count, as detailed in Table 7.

Table 7 demonstrates that MCTNet exhibited superior performance in the majority of mapping areas, underscoring the effectiveness of the proposed model in practical applications. However, in California's mapping area 2, LTAE surpassed MCTNet. Additionally, in California's mapping area 1, RF achieved greater accuracy compared to GL-TAE and LTAE. These results contrast with the classification accuracy results based on testing data presented in Table 5. A possible explanation for this discrepancy is the random selection of samples from the CDL. These samples may not be entirely representative of the real situation, leading to differences between the model's performance in actual mapping and on the testing data. A model's excellent performance on the testing data only suggests a higher likelihood of superior performance in actual

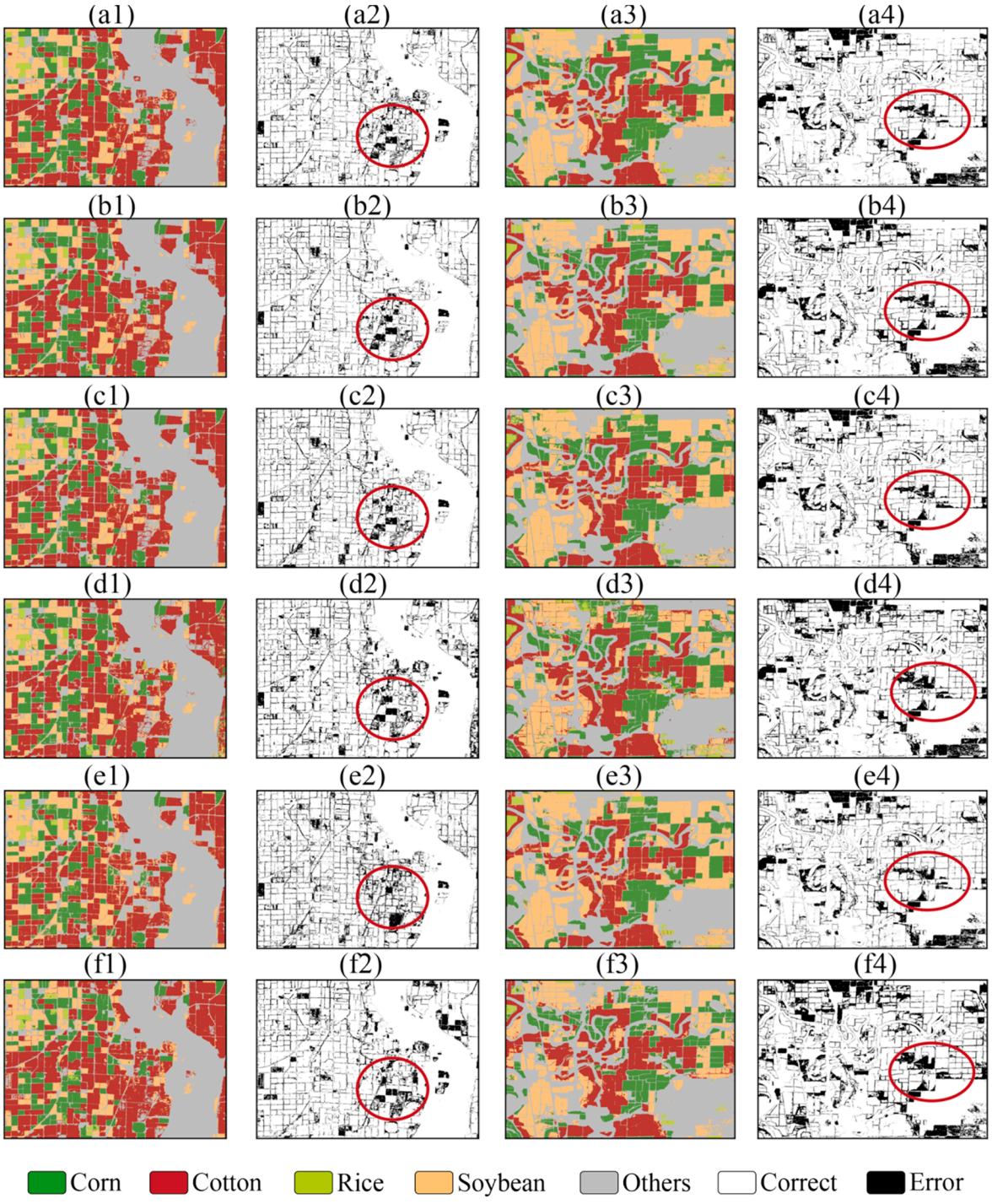

Fig. 9. Crop maps from (a) MCTNet, (b) GL-TAE, (c) LTAE, (d) ResNet, (e) Transformer, (f) RF in mapping area 1 (1) and mapping area 2 (3) in Arkansas. (2) and (4) represent the residual maps in mapping area 1 and mapping area 2, respectively.

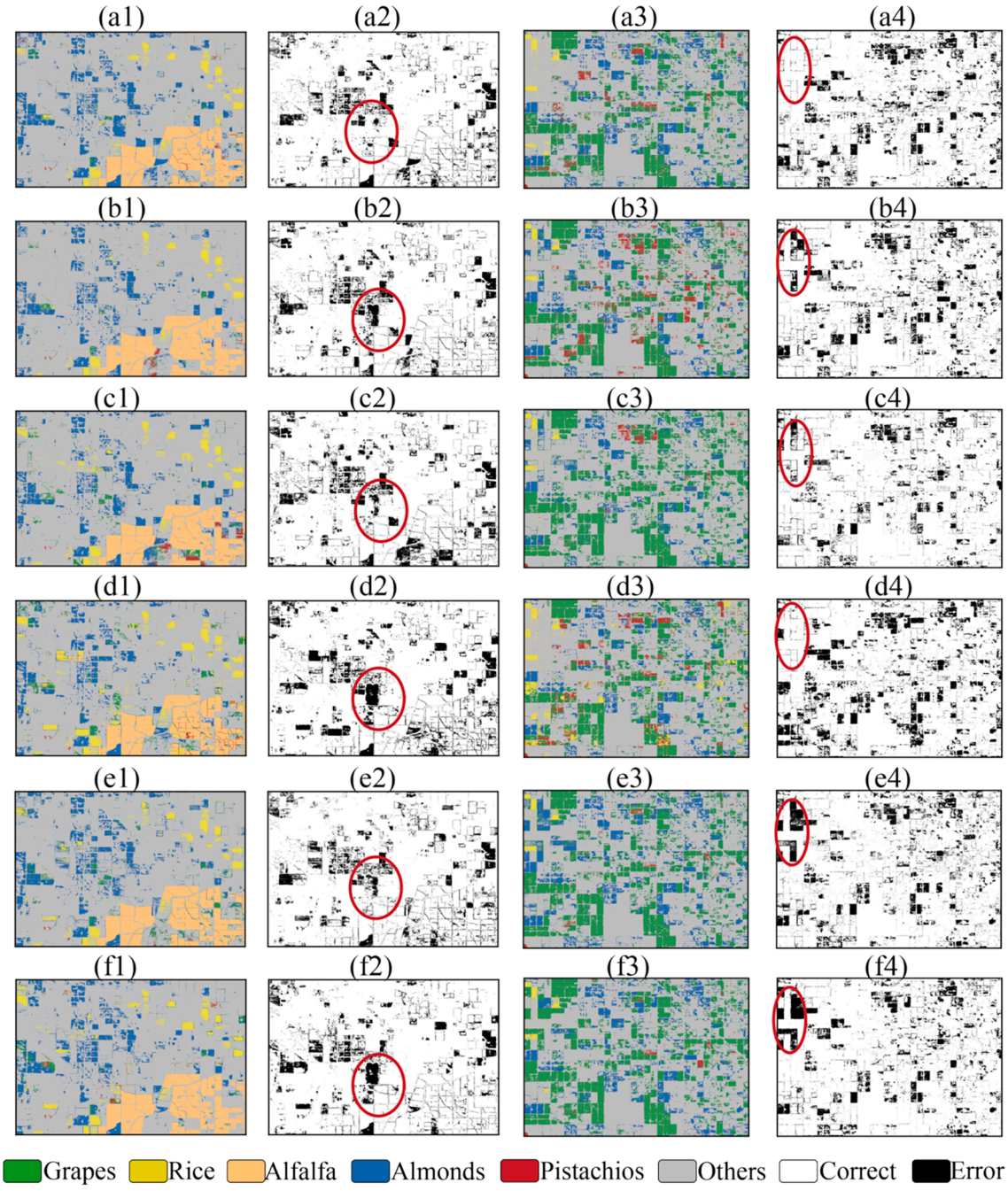

Fig. 10. Crop maps from (a) MCTNet, (b) GL-TAE, (c) LTAE, (d) ResNet, (e) Transformer, (f) RF in mapping area 1 (1) and mapping area 2 (3) in California. (2) and (4) represent the residual maps in mapping area 1 and mapping area 2, respectively.

Table 7
The mapping accuracy of different models.

State	Mapping Area	MCTNet	GL-TAE	LTAE	ResNet	Transformer	RF
Arkansas	1	**0.893**	0.888	0.883	0.871	0.877	0.875
	2	**0.854**	0.847	0.850	0.827	0.852	0.811
California	1	**0.897**	0.887	0.886	0.872	0.893	0.895
	2	0.873	0.871	**0.892**	0.851	0.868	0.869

mapping compared to suboptimal models, but it does not offer a guaranteed assurance of superior performance.

A detailed analysis of the predicted map of Arkansas (Fig. 9) shows that all models perform well except for the central region of map area 1. However, some pixels that should have been classified as cotton and other crops were inaccurately identified as soybeans. In mapping area 2, where cotton, soybean, and corn are predominant, all models reflected the overall crop distribution pattern. MCTNet demonstrated superior accuracy with fewer misclassifications, particularly in the central area. Fig. 10 demonstrates that the crop fields in California are much smaller

and more fragmented than those in Arkansas. In mapping area 1, all predicted maps identified the alfalfa correctly but failed to map the other crops very accurately. Mapping area 2 is planted with many grapes, along with scattered almonds, pistachios, and rice. The mapping results of mapping area 2 were similar to those of mapping area 1, with high recognition accuracy for widely planted crops (grapes) but low accuracy for small-scale crops. Notably, MCTNet yielded fewer errors in certain localized areas. Nonetheless, all models produced more salt-and-pepper errors in California than in Arkansas, which may be attributed to fragmented and heterogeneous landscapes.

3.4. Cross-temporal performance

In real applications, it is often impossible to collect training data for specific years, making the evaluation of the transferability and generalizability of models crucial. To explore cross-temporal performance, we conducted an additional experiment to assess the performance of models across different years. Specifically, we collected 5000 samples from each study area in 2020 and 2022, and tested the performance of models trained on 2021 data using these new datasets. The results, shown in Fig. 11, indicate that the proposed MCTNet consistently outperforms other models, achieving the highest scores in most scenarios, particularly in Arkansas 2020 and 2022, and California 2020. Although MCTNet ranks second to RF in California 2022, it remains highly competitive. In contrast, the comparison models exhibit significant performance variations across different scenarios. For example, while the RF model maintains good performance across years in California, it shows inferior performance in Arkansas. This indicates that the proposed model effectively and precisely captures the temporal patterns of crops, demonstrating superior robustness and stability in handling unseen temporal data.

4. Discussion

4.1. Model interpretation

The results in Section 3 substantiated the efficacy of the MCTNet by leveraging the integration of the CNN and the Transformer to enhance feature extraction capabilities. To further investigate the contribution of the CNN and Transformer sub-modules in the proposed model, the gradient-weighted class activation mapping (Grad-CAM) (Selvaraju et al., 2017) was applied to visualize the extracted features by the sub-modules. Specifically, the Grad-CAM on the last convolutional layer of the CNN sub-module and the last normalized layer of the Transformer sub-module were selected for visualization. To explore the effects of the multiple stages in the MCTNet, the visualization results were also produced for each stage.

Fig. 12 (a1) – (a3) demonstrate the Grad-CAM results of CNN sub-modules using the corn samples in Arkansas. In the spectral dimension, the red edge and NIR bands, known to be sensitive to the physiological state of plants (Sharifi, 2020), exhibit significant importance. In the temporal domain, the features near the 180 DOY obtain more attention. This period corresponds to the tasseling and silking stages of corn, during which the plants undergo significant structural and color changes. From a remote sensing perspective, these developmental stages are characterized by notable alterations in the spectral reflectance of the corn canopy, making them easily detectable. Additionally, as illustrated in Fig. 2, the occurrence of these stages in corn is significantly earlier compared to other crops. Consequently, the spectral features during these stages are important for crop identification and thus receive heightened attention. Moreover, we found that the important features become more concentrated from the first stage to the last stage. This is because the features learned by deep CNN models exhibit hierarchical characteristics, that is, the shallow layers learn abstract features while the deeper layers learn specific and invariant features that serve for classification (Zeiler and Fergus, 2013). Fig. 12 (c1) – (c3) also shows a gradual concentration phenomenon in California with grape samples. Different from corn, the important features of grapes are a little bit away from the peak value date (Fig. 2). Deep learning models are often viewed as 'black boxes' with limited interpretability, and the optimal combination of features might not include the peak date, especially when the peak is not markedly evident. In contrast, the Grad-CAM results of Transformer sub-modules (Fig. 12 (b1) – (b3) and Fig. 12 (d1) – (d3)) are different from those of the CNN sub-modules. The important features of Transformer sub-modules become more scattered along the temporal domain from the first stage to the last stage. This is because CNN models

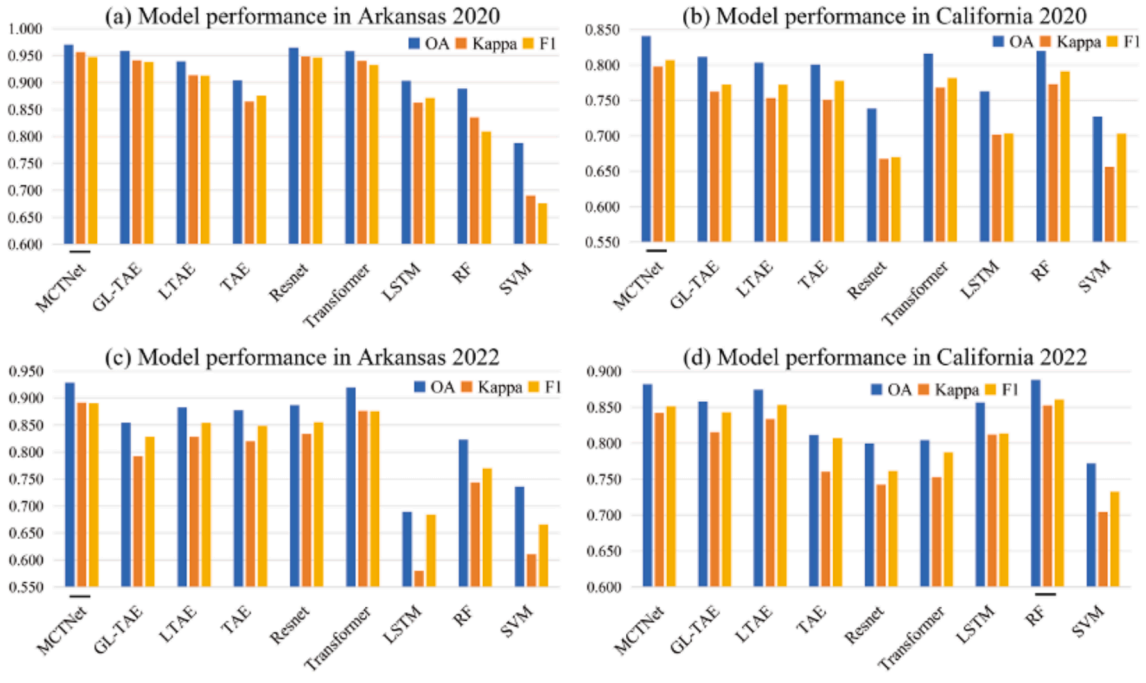

Fig. 11. Cross-temporal performance of different models ("_" represents the best model).

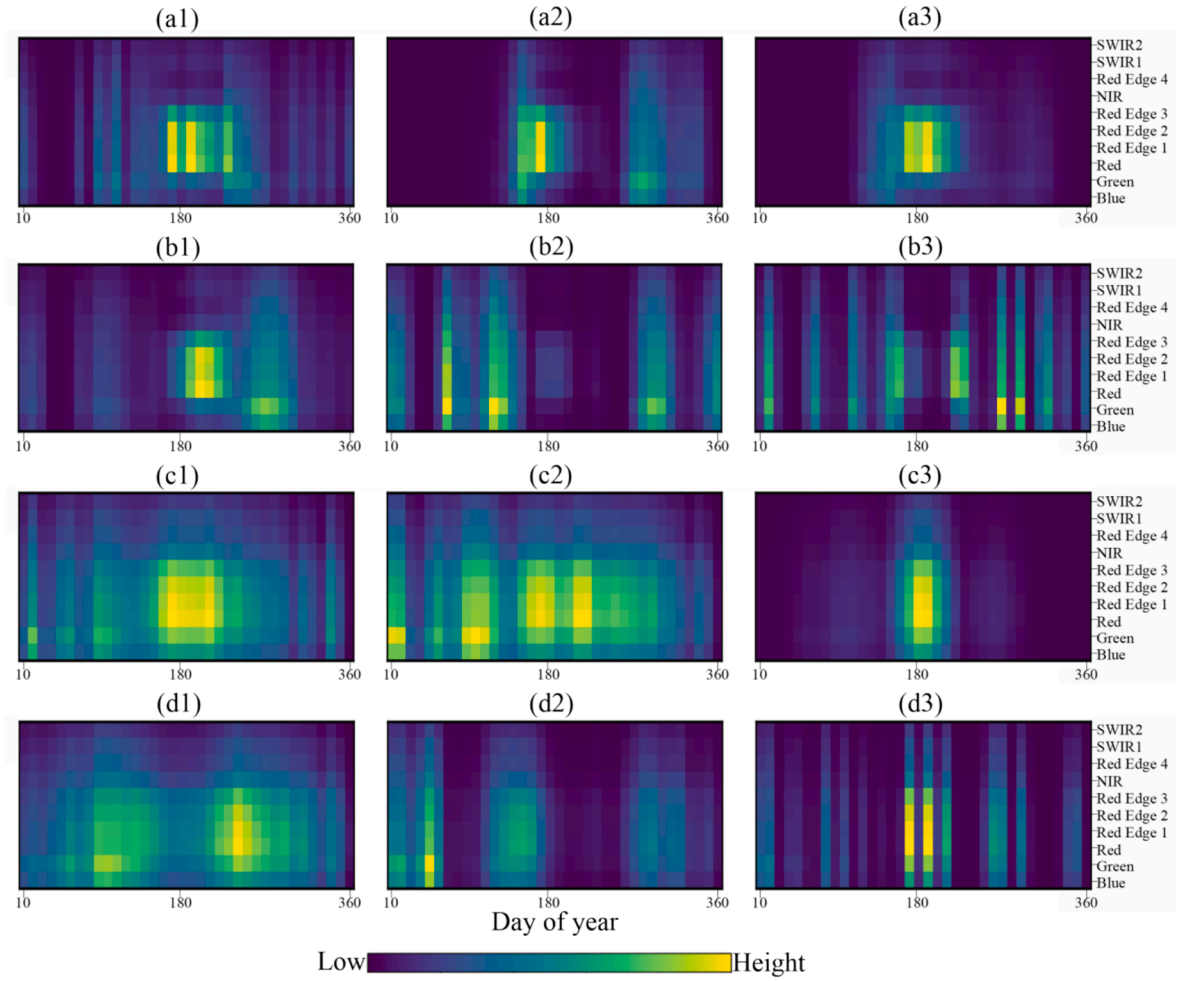

Fig. 12. Grad-CAM visualization results for (a) corn in Arkansas by CNN sub-module, (b) corn in Arkansas by Transformer sub-module, (c) grape in California by CNN sub-module, (d) grape in California by Transformer sub-module. (1) to (3) represent the tree stages in MCTNet. Yellow color indicates high-weight features, while blue color means low-weight features. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

focus on local features while Transformer models can capture long-range feature dependencies. The Transformer sub-module can classify crops by learning a specific pattern formed by different temporal features. Meanwhile, we observed that the important features extracted by CNN and Transformer sub-modules exhibited certain complementarity, which can theoretically enhance the accuracy of crop classification.

To further investigate how the extracted features affect the classification, t-SNE (Van der Maaten and Hinton, 2008), a widely used dimensionality reduction approach, was applied to visualize the distributions of the extracted features of the sub-modules in each stage. Fig. 13 (a1) – Fig. 13 (a3) demonstrates the visualized results of CNN sub-modules in Arkansas for three stages, respectively. Obviously, in the first stages, the distribution of the extracted features is highly concentrated and forms several indistinguishable clusters. In the subsequent stages, these clusters gradually aggregate into a few large groups, with each large group representing a type of crop. The Silhouette scores also prove that the clustering performance becomes better in the latter stages. The visualized plots of Transformer sub-modules are similar to those of the CNN sub-modules. Slightly differently, the Transformer sub-modules form more and smaller clusters in the first stage. Fig. 14 displays similar visualized results in California, which further demonstrate the superiority of the multiple stages in improving crop discrimination.

4.2. The impact of missing data on model performance

To investigate the impact of varying degrees of missingness in time-series data on model accuracy, we calculated the missing rates for each sample in the testing dataset in Arkansas and then assessed the robustness of MCTNet and MCTNet_noALPE models using samples with missing rates less than 10 %, 15 %, 20 %, 25 %, 30 %, 35 %, and 40 % respectively. We believe it is important to maintain realistic observations, so we did not refer to previous studies (Wang et al., 2022b; You et al., 2021) using linear interpolation and Savitzky-Golay smoothing to process time-series data to eliminate missing values. To verify whether these steps are necessary, we used the data that underwent interpolation and smoothing to generate two new models (named MCTNet1 and MCTNet_noALPE1). The model performance and the distribution of missing rates of data are presented in Fig. 15. It is obvious that the data with less than 10 % missingness constituted less than 20 % of the total, which implies that the vast majority of the data contained missing values with different rates. As the missing rate increased, the accuracy of all models gradually declined. The MCTNet achieved the best performance at every rate, and its OA decreased slightly, from 0.973 to 0.968. By contrast, the OA of MCTNet_noALPE dropped from 0.963 to 0.950. This demonstrates that MCTNet, owing to the contribution of ALPE, was less sensitive to data missingness than MCTNet_noALPE.

It is noteworthy that MCTNet significantly outperformed MCTNet1 across all missing scenarios, indicating that interpolation and smoothing might alter the original distribution of the data, leading to a reduction in model accuracy. In contrast, the accuracy of MCTNet_noALPE1 was persistently higher than that of MCTNet_noALPE, especially when data missingness is substantial. This discrepancy arose because

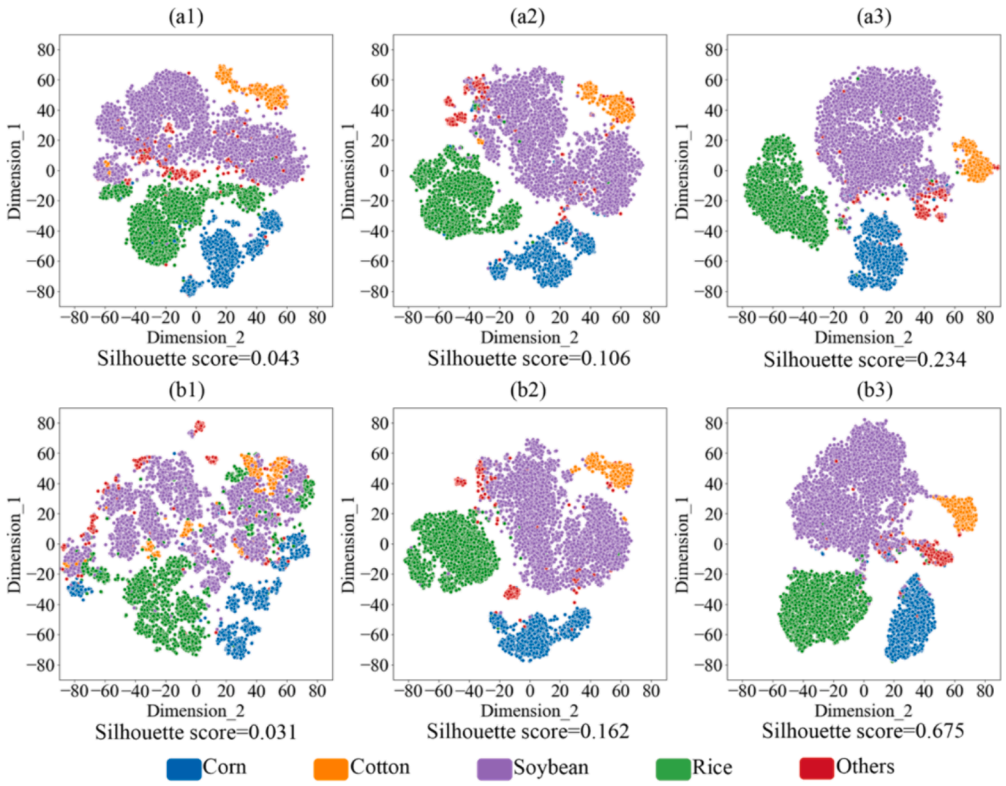

Fig. 13. The extracted feature distributions of (a) CNN sub-module, (b) Transformer sub-module from (1) the first stage to (3) the last stage in Arkansas.

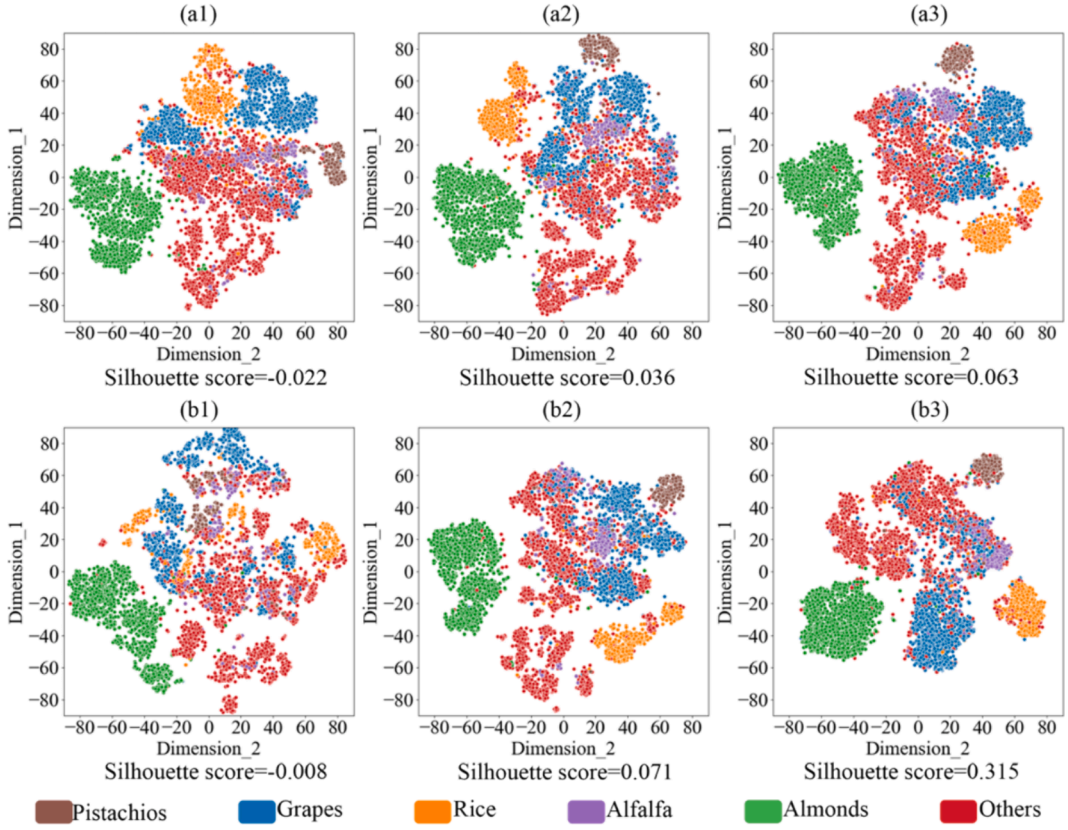

Fig. 14. The extracted feature distributions of (a) CNN sub-module, (b) Transformer sub-module from (1) the first stage to (3) the last stage in California.

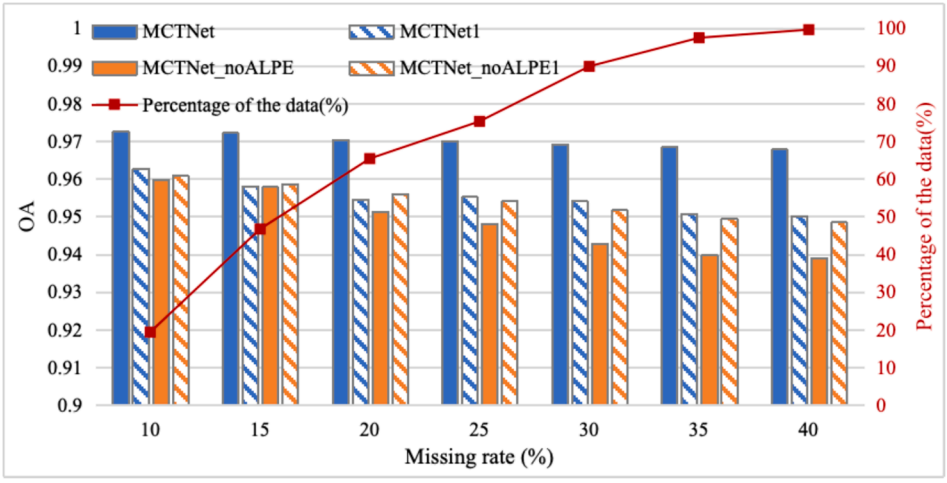

Fig. 15. The OA of the crop classification models using samples with different missing rates in Arkansas. (The red curve represents the percentage of the data that is less than a certain missing rate.). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

MCTNet_noALPE lacks the ALPE module to discern the locations of missing values. In cases where there are extensive missing values, the model becomes confused and cannot extract valid features to identify crops. The interpolation and smoothing can reduce many missing values and restore the original data distribution to a certain extent, thereby improving the recognition accuracy of the model. Overall, MCTNet is significantly superior to other models, and it does not require interpolation and smoothing, which can save a lot of calculations.

We further investigated the different impacts of random and continuous missing data during the growing season on model performance. Specifically, we used corn samples in Arkansas as an example. The growing season of corn is approximately from DOY 120 to DOY 270, as shown in Fig. 2. We collected all corn samples in Arkansas that had no missing data during this period, totaling 1,361 samples. For each sample, we created time series data with missing points by deleting a specific rate of the original data both randomly and consecutively. We then tested the processed samples using MCTNet and MCTNet_noALPE to assess the impacts of different missing conditions on the models' performance.

The results are shown in Fig. 16. As the missing rate increases, the OA decreases for all models. Notably, random missing data (solid bars) consistently results in higher performance compared to continuous missing data (striped bars). This is because random missing data are spread throughout the growing season, allowing the model to capture some information during critical growth stages. In contrast, continuous missing data creates prolonged gaps, causing the model to miss key phenological information and leading to a greater decline in performance. Additionally, MCTNet consistently outperforms MCTNet_noALPE across all missing data rates, demonstrating the ALPE module's effectiveness in improving model robustness against missing data.

To explore the impact of the ALPE in detail, we randomly selected a corn sample and extracted the corresponding attention weight matrices from the MCTNet and MCTNet_noALPE. These matrices are from the first head of the first Transformer sub-module of the models. We used bipartite graphs, a widely used visualization tool for such matrices, to display the attention distribution, where the strength of the attention is indicated by the opacity of the edges and nodes as described by Rußwurm and Körner (2020). Specifically, Fig. 17(a) plots the time series NIR values of the sample, indicating that the period between DOY 180 and DOY 300 represents the phenological period of the corn. However, the values for DOY 30, 40, 50, 60, 70, 110 and 360 are missing. Fig. 17 (b) and (c) show the attention distributions of MCTNet and MCTNet_noALPE, respectively. It is worth noting that the attention weights lower than 90 % were filtered out for enhanced clarity. The attention distribution of MCTNet demonstrates that the attentions are focused on the phenological period of the crop and are almost unaffected by the missing values. In contrast, the attention distribution of MCTNet_noALPE is more dispersed, with the attention being attracted

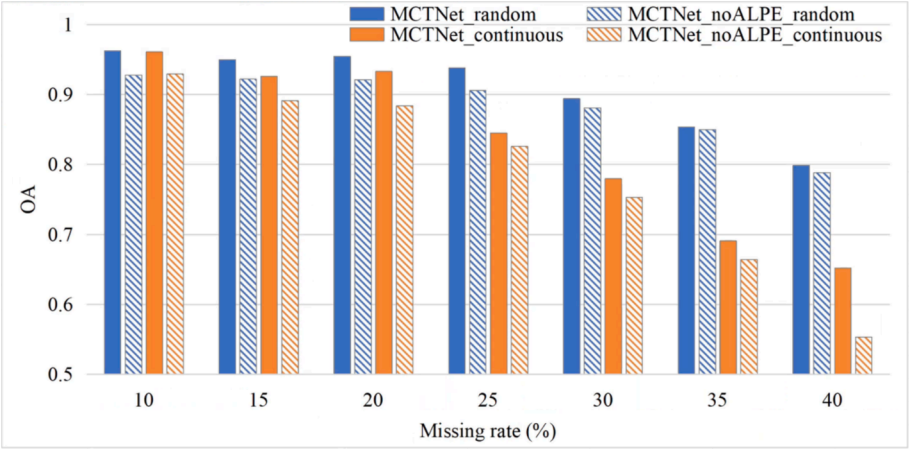

Fig. 16. Performance of the crop classification models using corn samples with different missing rates during the growing season ("random" denotes random missing data, "continuous" denotes continuous missing data).

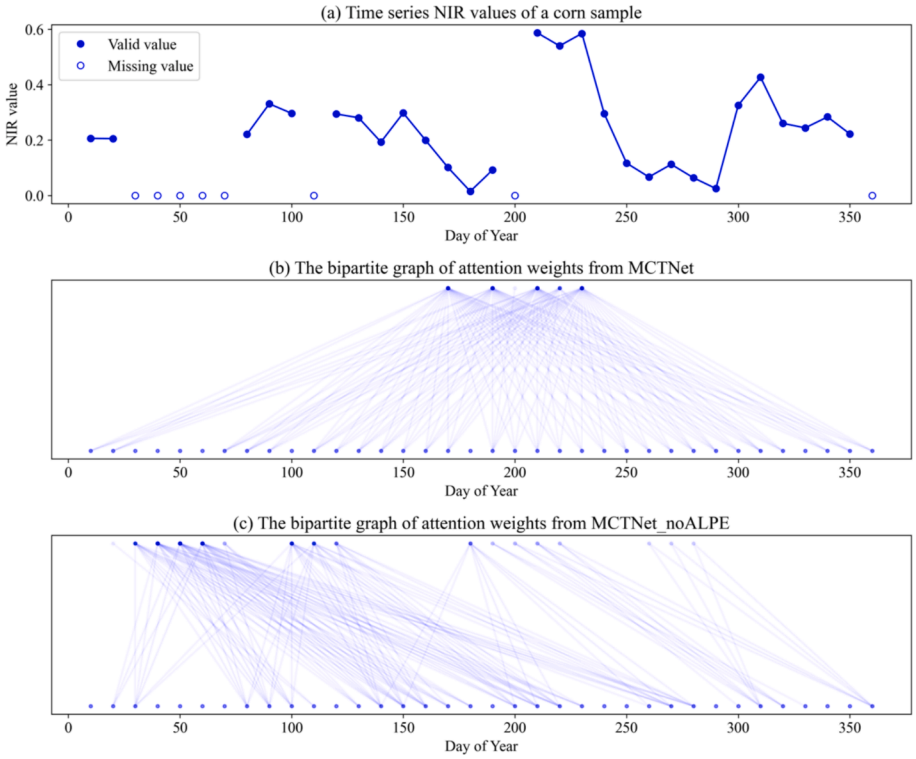

Fig. 17. Visual illustration of the attention weights from the MCTNet and MCTNet_noALPE.

by the missing values. This figure demonstrates the differences in attention distribution, highlighting how the ALPE module adjusts the attention weights to better handle missing data and enhance the model's ability to capture relevant temporal dependencies.

4.3. Limitations and future work

Although the proposed model achieved superior performance in crop mapping, there are several limitations for improvement. First, the MCTNet model may produce salt-and-pepper noise in classification results. This noise results from pixel-wise decision-making, which can be pronounced in heterogeneous, small, and broken agricultural landscapes (like California (Fig. 10)). Further studies will consider the spatial information of the input pixel to improve mapping accuracy. For example, as suggested by Interdonato et al. (2019), incorporating unlabeled pixels surrounding the input pixel in the modeling could enhance performance. Another limitation of our study is the exclusive use of optical data, while recent studies have shown the efficiency of SAR and hyperspectral data in crop monitoring and classification (Farmonov et al., 2023; Sharifi and Hosseingholizadeh, 2020). We plan to integrate multi-modal remote sensing data in future research to enhance the robustness and accuracy of crop type classification. Lastly, although the current study areas cover two U.S. states with unique agricultural characteristics, it remains imperative to explore the performance of the proposed model in various other countries and regions.

5. Conclusions

This study proposed a lightweight CNN-Transformer network MCTNet for pixel-based crop mapping using time-series Sentinel-2 images. MCTNet effectively harnesses the advantages of both CNN and Transformer to obtain a comprehensive representation of crop growth patterns. Compared with eight advanced models, the MCTNet exhibited better performance in most crop classification cases with the fewest number of parameters. The evaluation of the impact of data missing on the models demonstrated that the proposed ALPE made the MCTNet less sensitive to the missing values and achieved superior performance without data interpolation and smoothing. Moreover, MCTNet achieved superior performance in cross-temporal scenarios, indicating its high potential in practical applications. Further work aims to improve the model's productivity and generalizability across different regions.

CRediT authorship contribution statement

Yumiao Wang: Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Luwei Feng:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis. **Weiwei Sun:** Resources, Funding acquisition, Conceptualization. **Lihua Wang:** Investigation, Formal analysis, Data curation. **Gang Yang:** Visualization, Software, Funding acquisition. **Binjie Chen:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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